Area/delay

Mappers optimize different objectives: area, delay, or a blend

The idea

- Area_mode reports area + delay_sum; delay_mode reports depth×AND delay

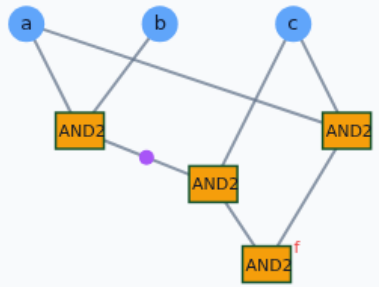
Trade

AREA/DELAY

Trade

Step 1 / 5 · trade · module03-03-area-delay

area 9



Explanation

Two modes share the same cover on this toy.

- Objectives

```
Report both
```

Trade

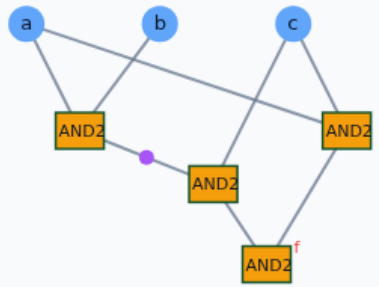
Area mode

AREA/DELAY

Area mode

Step 2 / 5 · area · module03-03-area-delay

area 9



Explanation

Sum of cell areas (plus INV).

- areaMode

```
Primary
```

Area mode

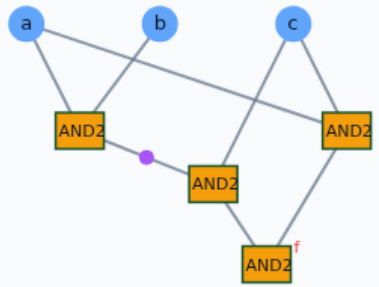
Delay mode

AREA/DELAY

Delay mode

Step 3 / 5 · delay · module03-03-area-delay

depth 8



Explanation

Depth proxy = #AND × AND2 delay.

- delayDepth

```
Proxy
```

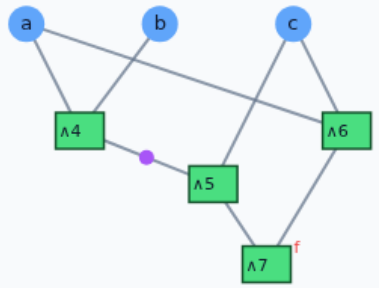
Delay mode

Library

AREA/DELAY

Library

Step 4 / 5 · lib · module03-03-area-delay



Explanation

Six cells bound the choice set.

- libCount

6

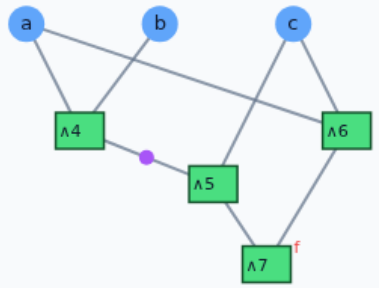
Library

Choice

AREA/DELAY

Choice

Step 5 / 5 · choice · module03-03-area-delay



Explanation

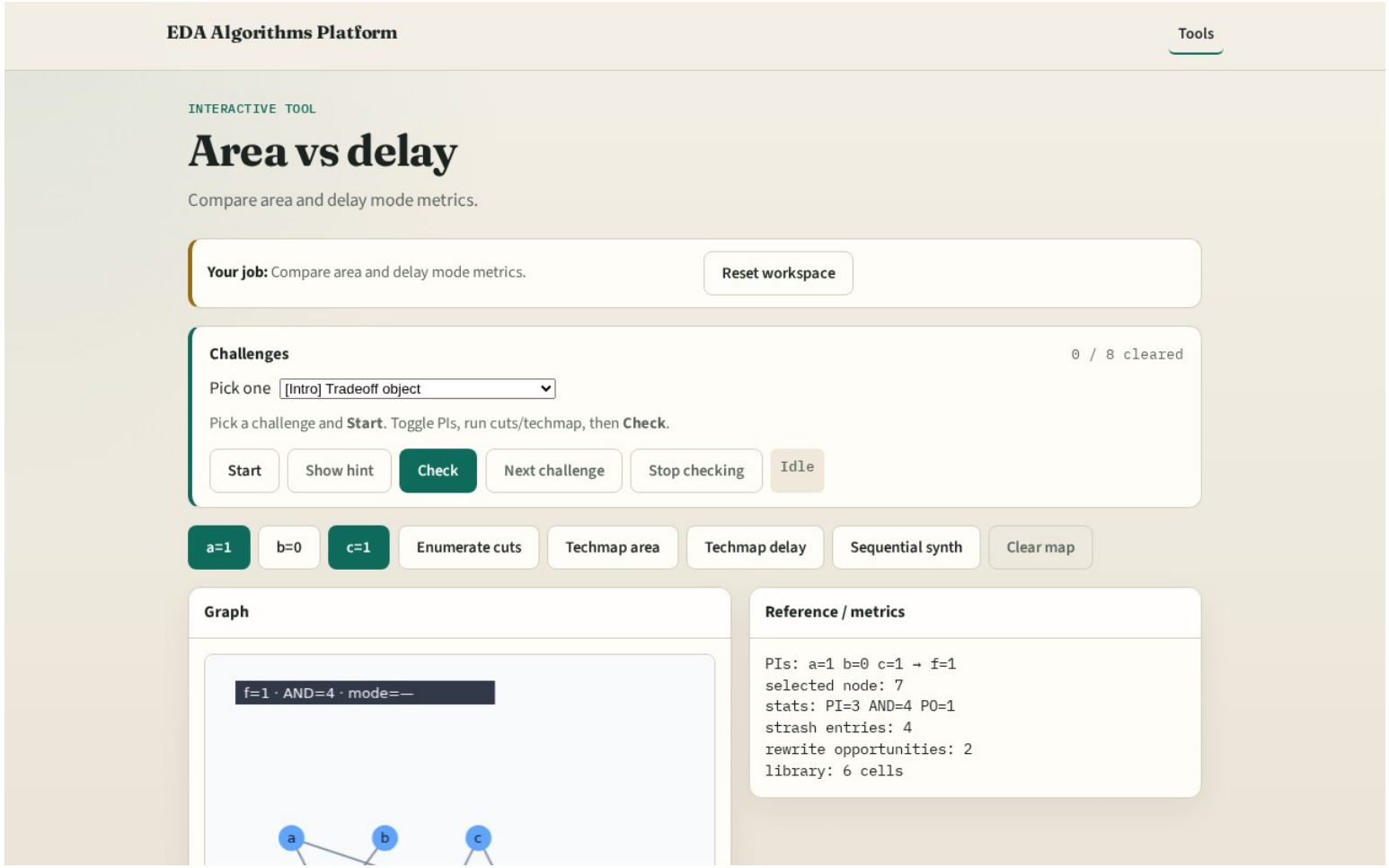
Real flows weight the two; we just report both.

- Literacy

```
Next seq
```

Choice

Browser lab track



Browser lab starter

Implement track

- Call area_delay_tradeoff; print both modes

Pitfalls

- Treating delay_sum as critical path blindly
- Ignoring depth

Your turn

- Next: full sequential synth pass